ORF 409 Midterm

(Q.1) Note that :-

$Q(t)$ = # people in the system = n = $N_A(t) - N_D(t)$

Earlier, we were mapping time to people (arrival & departure times of people).

Now, we map people to time (how many people at $t = 30.2$).

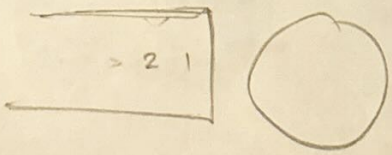
$\Rightarrow$ create fixed array of length 3000.

~~Suppose A[-1] = 100 $\Rightarrow$ all updates after~~

$\Rightarrow$ use A & D i.e. arrival & departures times of people to populate NA & ND (# people at each time instant)

Suppose $A[-1] = 100 \Rightarrow$ all updates after $t = 100$ will be with the same na ($\because$ no new arrival after this).

Now, how to use/process PS ?



$S(1) = 12$

$N(2) = 5$

$N(4)$

so tw $s = \frac{5}{12}$

After 5, if $N(3) = 15$

$\frac{N(4) - N(3)}{3}$, $\frac{N(5) - N(4)}{4}$

$\Delta = 10$ but $s(1)t = \frac{\Delta}{2}$

$\frac{N(k+1) - N(k)}{k}$

What happens in processor sharing?

How is td determined? → The issue is:- td is not fixed even if once determined. Why? Because as $n \uparrow$, td gets pushed behind

⇒ td dependent on n ⇒ Arrivals

However arrivals indp of departures ⇒ Arrivals still in $t_a < t_d$ loop but departures not completely outside it.

Now, once a job exits, why isn't td updated? This way, won't it be ∞ always?

Essentially, what happens to td when after departure, $n \neq 0$?

Some update needed when

$t_e == t_d$, 1 departs but still $n > 0$ ⇒ otherwise

$t_d = \min(t_d, t_a)$ & that will keep on continuing

↳ Added

(Q.2) X_n = negative Binomial Random variable with n dof

$= \text{Binomial}(p, -n)$; $0 < p < 1$

~~X_n = Binomial~~

X_n's support : $\{0, 1, 2, \ldots\} \Rightarrow X_n \geq 0$

(1) Note that :-

$$n \geq n-1$$

$$\Rightarrow n + X_n \geq n-1+X_n$$

$$\Rightarrow \frac{1}{n+X_n} \leq \frac{1}{n-1+X_n} \quad (n \geq 2 \Rightarrow n-1 \geq 1)$$

$$\Rightarrow \frac{X_n}{n+X_n} \leq \frac{X_n}{n-1+X_n} \quad (\because X_n \geq 0)$$

$$\Rightarrow 1 - \frac{X_n}{n+X_n} \geq 1 - \frac{X_n}{n-1+X_n}$$

$$\Rightarrow \frac{n+\cancel{X_n}-\cancel{X_n}}{n+X_n} \geq \frac{n-1+\cancel{X_n}-\cancel{X_n}}{n-1+X_n}$$

$$\Rightarrow \frac{n}{n+X_n} \geq \frac{n-1}{n-1+X_n}$$

– QED

Now, $\text{pmf}(X_n) = P(X_n = k) = {}^{k+n-1}C_k \, p^{n-1}(1-p)^k$

So, $E\left(\frac{n-1}{n-1+X_n}\right) = \sum_{k=0}^{\infty} P(X_n = k)\left(\frac{n-1}{n-1+k}\right)$

$$= \sum_{k=0}^{\infty} {}^{k+n-1}C_k \, p^{n-1}(1-p)^k \left(\frac{n-1}{n-1+k}\right) = \sum_{k=0}^{\infty} \frac{(k+n-1)!}{k!\,(n-1)!} p^{n-1}(1-p)^k \left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} \frac{(k+n-2)!}{k!\,(n-2)!} p\, p^{n-2}(1-p)^k = p$$

$X \sim Bin(p, -n) \Rightarrow P(X_n = k) = {}^{-n}C_k \, p^k (1-p)^{-n-k}$

$= {}^{k+n-1}C_k \, p^k (1-p)^n$ $\quad$ ~~$= {}^{n+k-1}C_k$~~

!

$(p = \text{prob of } \cancel{\text{failure}}\ \text{success})$ $\rightarrow$ we want n losses here (keep rolling till we have n losses)

$$\Rightarrow E\left(\frac{n-1}{n-1+X_n}\right) = \sum_{k=0}^{\infty} P(X_n = k)\left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} {}^{k+n-1}C_k \, p^k (1-p)^n \left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} \frac{(k+n-1)!}{k!\,(n-1)!} p^k (1-p)^n \left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} \frac{(k+n-2)!}{k!\,(n-2)!} p^k (1-p)^{n-1} (1-p)$$

$$= (1-p) \sum_{k=0}^{\infty} {}^{k+n-2}C_k \, p^k (1-p)^{n-1}$$

$$= (1-p) \sum_{k=0}^{\infty} P(Bin(p, -(n-1)) = k$$

$$= \underline{\underline{1-p}}$$

$\hookrightarrow$ still a Negative Binomial RV

$\because n \geq 2 \Rightarrow n-1 \geq 1 > 0$

— QED

(2) Let 'n' degrees of freedom correspond to events of type A (1)

$\Rightarrow$ The other events (that X_n counts for) = type B. (2)

So, p = prob (event B occurs).

~~To prove: $\lim_{n\to\infty} \frac{n}{n+X_n} = \lim_{n\to\infty} \frac{n-1}{n-1+X_n} = 1-p$~~

~~Proof: Note that $\frac{n}{n+X_n} = \frac{A}{A+B} = P(A) = 1-P(B) = 1-p$~~

~~$\Rightarrow$ To simulate X_n (with this thing), we can invert simulation so that now since $n \gg X_n$, we simulate 'n' instead of X_n (but not normally but instead using the ~~Gaussian~~ Geometric technique $\Rightarrow$ If V is Geometric RV, V represents # events that will happen before we get an event A).~~

~~So, if V = 5, put 5 '2' and then a '1' & keep doing so (keep simulating for r) until we have n '1's~~

Note that X_n represents total # successes till we have n failures

So, $\frac{n}{n+X_n} \sim \frac{\#f}{\#s+\#f} = 1 - \frac{\#s}{\#s+\#f} \xrightarrow{n\to\infty} 1-p$

As $n \to \infty$, $n-1 \to n \Rightarrow n-1+X_n \to n+X_n \Rightarrow \frac{1}{n-1+X_n} \to \frac{1}{n+X_n}$

$\Rightarrow \frac{n-1}{n-1+X_n} \to \frac{n}{n+X_n} \Rightarrow \lim_{n\to\infty} \frac{n-1}{n-1+X_n} = \lim_{n\to\infty} \frac{n}{n+X_n}$.

& $X_n = \sum Z_i$ ($Z_i \to$ i) Random
ii) Ind^p
iii) larger n
iv) $var(Z_i)$ = finite $\to \frac{p}{(1-p)^2}$
v) $E(Z_i) = \frac{p}{1-p}$ = finite

$\Rightarrow$ By ~~CLT~~ WLLN,

$\frac{1}{n}\sum Z_i \xrightarrow{n\to\infty} E(Z_i) = \frac{p}{1-p}$

$\Rightarrow \lim_{n\to\infty} \frac{X_n}{n} = \frac{p}{1-p}$

$\Rightarrow \lim_{n\to\infty} 1 + \frac{X_n}{n} = 1 + \frac{p}{1-p}$

$\Rightarrow \lim_{n\to\infty} \frac{n+X_n}{n} = \frac{1}{1-p}$

$\Rightarrow \lim_{n\to\infty} \frac{n}{n+X_n} = 1-p$ — (1)

How to simulate X_n?

Note that X_n is nothing but the sum of n iid geometric RV, each with parameter p (considering conventions as described in the class)

$$\Rightarrow \quad X_n = \sum_{i=1}^{n} Z_i \quad \text{where } Z_i \sim \text{Geometric}(p) \text{ iid.}$$

Now, each Z_i can be simulated using :-

M1 : uniform for each trial

M2 : getting a number representing # trials before the (first) failure occurs

M3 : Problem 3, part 1 of the midterm

I used M3 as of now.

once we have realizations of Z_i, say z_i.

$$x_n = \sum_{i=1}^{n} z_i$$

Realization of X_n

$Z_i =$	0	1	2	3	...
w.p.	$(1-p)$	$p(1-p)$	$p^2(1-p)$	$p^3(1-p)$	...

$$E(Z_i) = 0(1-p)p + 2(1-p)p^2 + 3(1-p)p^3 + \dots$$

$$p\,E(Z_i) = (1-p)p^2 + 2(1-p)p^3 + \dots$$

$$(1-p)\,E(Z_i) = p(1-p) + p^2(1-p) + \dots$$

$$= \frac{p(1-p)}{1-p} = p \Rightarrow E(Z_i) = \frac{p}{1-p}$$

(Q.3) $\log_e x$, $0<p<1$, $U \in U[0,1]$

① To prove :- $\lfloor \log_{1/p}(1/U) \rfloor$ = Geometric (p) RV

Proof :- Note that :-

$$\lfloor \log_{1/p}(1/U) \rfloor \le \log_{1/p}(1/U) < \lfloor \log_{1/p} 1/U \rfloor + 1 \quad - ①$$

$$\Rightarrow \log_{1/p}(1/U) - 1 < \lfloor \log_{1/p}(1/U) \rfloor \le \log_{1/p}(1/U) \quad - ②$$

let $\lfloor \log_{1/p}(1/U) \rfloor = X = \alpha$

$$\Rightarrow P(X=\alpha) = P\left(\alpha \le \log_{1/p}(1/U) < \alpha+1\right) \quad (\text{from } ①)$$

$$= P\left(\left(\tfrac{1}{p}\right)^{\alpha} \le \tfrac{1}{U} < \left(\tfrac{1}{p}\right)^{\alpha+1}\right) \quad (\because p<1,\ \tfrac{1}{p}>1)$$

$$= P\left(p^{\alpha} \ge U > p^{\alpha+1}\right)$$

$$= P\left(p^{\alpha+1} < U \le p^{\alpha}\right)$$

$$= p^{\alpha} - p^{\alpha+1}$$

$$= p^{\alpha}(1-p)$$

$\Rightarrow$ pmf of X matches that of Geometric R.V with p = probability (success) (as per class notation)

Also, $\log_{1/p} 1/U = \log_{>1} >1 \ge 0 \Rightarrow X \ge 0 \Rightarrow \alpha \ge 0 \ (\forall \alpha \in \mathbb{Z}_{\ge 0})$

$\Rightarrow$ support set (X) = support set (Geometric RV)

↳ (As discussed in class, this is 0, 1, 2, ...)

$\Rightarrow X = \lfloor \log_{\frac{1}{p}}(\tfrac{1}{U}) \rfloor$ = Geometric RV $\Rightarrow$ QED

(Q.4) price of the stock:- $S(t) = S(0)\exp(\sigma W(t) + (\mu - \frac{\sigma^2}{2})t)$

$\cancel{\Rightarrow S(t+\Delta t) = S(t)\exp(\sigma(W(t+\Delta t) - W(t)) + (\mu - \frac{\sigma^2}{2})\Delta t)}$

$$S(t+\Delta t) = S(0)\exp(\sigma W(t+\Delta t) + (\mu - \frac{\sigma^2}{2})(t+\Delta t))$$

$$\Rightarrow \frac{S(t+\Delta t)}{S(t)} = \frac{\cancel{S(0)}\exp(\sigma W(t+\Delta t) + (\mu - \frac{\sigma^2}{2})(t+\Delta t))}{\cancel{S(0)}\exp(\sigma W(t) + (\mu - \frac{\sigma^2}{2})t)}$$

$$\Rightarrow \frac{S(t+\Delta t)}{S(t)} = \exp(\sigma(W(t+\Delta t) - W(t)) + (\mu - \frac{\sigma^2}{2})\Delta t)$$

$\cancel{\Rightarrow S(t+\Delta t)}$

$$\Rightarrow \boxed{S(t+\Delta t) = S(t)\exp(\sigma(W(t+\Delta t) - W(t)) + (\mu - \frac{\sigma^2}{2})\Delta t)}$$

To be used for simulation

~~step1 : multiple~~

(Q.5)

$a(i, j) \longrightarrow$ # people in the queue after this event

$\downarrow$

people currently in the queue

If $j = i+1$,

$i \longrightarrow i+1 \Rightarrow$ arrival $\Rightarrow \lambda$

elif $j = i-1$,

$i \longrightarrow i-1 \Rightarrow$ departure $\Rightarrow \mu$ (service completed)

What we expect :-

If $\lambda > \mu$, $n \uparrow$

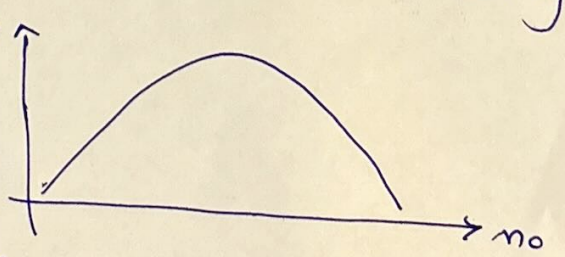
$\lambda < \mu$, $n \downarrow$

So, $n_0 = \underbrace{2 \quad 6}_{\lambda < \mu (n \to 0)} \quad 10 \quad \underbrace{14 \quad 18}_{\lambda > \mu (n \to 20)}$

$\lambda < \mu (n \to 0) \rightarrow$ t early

$\lambda > \mu (n \to 20) \rightarrow$ t early

① when $\lambda (= \mu) = 1 \longrightarrow$

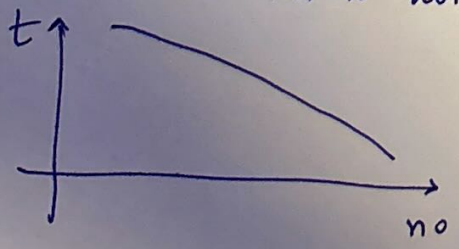


② $(2.5)\ \lambda > \mu = 1$

$\Rightarrow$ more arrivals

$\Rightarrow$ n tends to $\uparrow$ $\Rightarrow$ tendency to reach 20

$\Rightarrow$ t less when no more

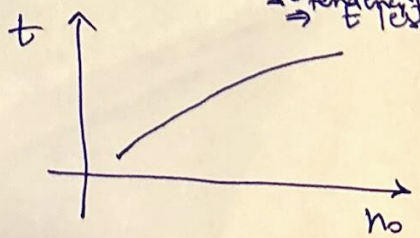


③ $\lambda (0.4) < \mu = 1 \Rightarrow$ more departures

$\Rightarrow$ n tends to $\downarrow$

$\Rightarrow$ tendency to reach 0

$\Rightarrow$ t less when no less



Also, $t \propto \frac{1}{\text{max rate}} = \frac{1}{\max(\lambda, \mu)}$

$$\Rightarrow \quad \begin{array}{lccc} \lambda = & 0.4 & 1 & 2.5 \\ \mu = & 1 & 1 & 1 \\ \max(\lambda, \mu) = & 1 & 1 & 2.5 \\ & t_1 & t_2 & t_3 \end{array}$$

for $n_0 = 10$, t_2 high in general.

for $n_0 = 2, 6$, t_3 high —//—

$n_0 = 14, 18$, t_1 —//—

However, t_3 for $\underbrace{n_0 = 2, 6}_{r_{max} = 2.5}$ $<$ t_1 for $\underbrace{n_0 = 14, 18}_{r_{max} = 1}$

because of rates